



7488 - Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at $z=7.3$

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	go7488_p1_g395m	NIRSpec MultiObject Spectroscopy	(2) primer-uds_imgv7.2_aper8_260528

ABSTRACT

The recent discovery of a massive quiescent galaxy at $z=7.3$, RUBIES-UDS-QG-z7, has pushed the most distant known such galaxy ~ 500 Myr closer to the Big Bang. Its mere existence challenges models and simulations which predict number densities that are ~ 100 times lower than implied by RUBIES-UDS-QG-z7, and struggle to reproduce its formation pathway.

While the available PRISM spectrum shows no emission lines and a strong Balmer Break, absorption features are only tentatively detected. Deeper, medium resolution spectroscopy is needed to confirm the quiescence of the source through direct detection of individual (Balmer) absorption lines. This will further enable the first ever measurement of a stellar velocity dispersion at $z > 7$, providing an independent probe of the total mass of the system, and yield much tighter constraints on the star formation history. These information are pivotal to gain confidence in the nature of the source, to better understand how it formed and quenched so rapidly so early on in the Universe, and to provide valuable constraints for models and simulations of high redshift galaxies.

OBSERVING DESCRIPTION

The main goals of this proposal are to detect and quantify Balmer absorption lines in RUBIES-UDS-QG-z7, as well as to infer its stellar velocity dispersion, σ_* from fitting the spectrum.

We choose the G395M/F290LP grating/filter pair with a 3 shutters slitlet shape. We use two different dithers/masks to mitigate cosmic rays with 3 nods each for optimal background subtraction, 5 integrations per exposure and 17 groups per integration with the NRSIRS2 readout, yielding a total exposure time of 37'639.34 s.

We will make sure that the primary target is well-centered in the MSA shutter to avoid unnecessary slit losses which will be minimal due to the compact size of the target. Additionally, the MSA will be filled with potential neighbors at $7 < z_{\text{phot}} < 7.6$, and any galaxies at $z > 7$.

Proposal 7488 - Targets - Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at $z=7.3$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	primer- uds_imgv7.2_aper8_260528	RA: 02 17 45.0543 (34.4377262d) Dec: -05 07 45.00 (-5.12917d) Equinox: J2000		
	Comments: Description=[]				

Proposal 7488 - Observation 1 - Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at z=7.3

Observation	Proposal 7488, Observation 1: go7488_p1_g395m										Thu Jun 18 12:00:24 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(go7488_p1_g395m (Obs 1)) Warning (Form): Config c1 (#1) has 1 primary slit traces affected by failed open shutters.										
	(go7488_p1_g395m (Obs 1)) Warning (Form): Config c1 (#2) has 1 primary slit traces affected by failed open shutters.										
	(go7488_p1_g395m (Obs 1)) Warning (Form): Config c1 (#3) has 1 primary slit traces affected by failed open shutters.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	primer-uds_imgv7.2_aper8_260528	RA: 02 17 45.0543 (34.4377262d) Dec: -05 07 45.00 (-5.12917d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	primer-uds_imgv7.2_aper8_260528 (30128 sources)			jwst-nirspec-g395m		1.5
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	110118	34.440817	-5.162878	24.629663467407227	1	174381	34.441294	-5.109215	24.147680282592773	
	1	114607	34.427481	-5.158003	24.562036514282227	1	175092	34.447071	-5.109977	24.62972068786621	
	1	139607	34.412767	-5.126395	23.887142181396484	1	176197	34.421652	-5.111622	23.852800369262695	
	1	151463	34.456244	-5.122505	24.459856033325195	1	181687	34.455472	-5.120375	23.89898109436035	
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time	
	1	c1		NRSIRS2RAPID	5	1		1		87.533	

Proposal 7488 - Observation 1 - Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at $z=7.3$

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395M/F290LP)	c1	3 Shutter Slitlet	34.434935541666 67 Degrees - 5.135023333333 22 Degrees	38.611008180273 07			3	9	12604.801
	2	1 (G395M/F290LP)	c1	3 Shutter Slitlet	34.434935541666 67 Degrees - 5.135023333333 22 Degrees	38.611007376380 13		0.1	3	9	12604.801
	3	1 (G395M/F290LP)	c1	3 Shutter Slitlet	34.434935541666 67 Degrees - 5.135023333333 22 Degrees	38.611006572494 21		0.2	3	9	12604.801
Special Requirements	Background Limited. Background no more than 50th percentile above minimum MSA Scheduled Aperture PA 38.6108 to 38.6108 Degrees (V3 260.0362 to 260.0362)										